Linux-eko Prozesuen Kudeaketa: Komandoen Azalpen Zehatza

Gida honetan erabilitako komando bakoitzak zehazki zer egiten duen azaltzen da, pausoz pauso. Zure dokumentuan komando bakoitzaren ondoan dagokion argazkia jarri ahal izateko antolatuta dago.

1. ps (Process Status)

- **Zer egiten du:** Sistema eragilean une horretan martxan dauden prozesuen argazki edo zerrenda bat erakusten du.
- **Nola erabili eta aldaerak:**
  - ps: Terminal horri lotutako prozesuak soilik erakusten ditu (adibidez, bertan dabiltzan shell-a eta komandoa bera).
  - ps aux: Sistemako erabiltzaile guztien prozesu guztiak zerrendatzen ditu xehetasun osoz (%CPU, %MEM, PID, estatua...).
  - ps axjf: Prozesuak zuhaitz formatuan eta hierarkian erakusten ditu, zein prozesuk (gurasoak) zein sortu duen (semeak) argi ikusteko.
  - ps aux | grep [izena]: Aurreko zerrenda iragazki baten bidez pasatzen du (grep), izen zehatz bat duen prozesu bat soilik bilatzeko.

2. top

- **Zer egiten du:** Denbora errealean eta modu dinamikoan sistemaren egoera eta prozesu kontsumitzaileenak erakusten dituen pantaila interaktiboa da.
- **Nola erabili eta aldaerak:**
  - top: Interfaze nagusia irekitzen du CPU, memoria eta prozesuen zerrendarekin. Hemendik irteteko q sakatu behar da.
  - top -d 5: Pantaila zenbat segundoz behin berritzen den zehazten du (kasu honetan, 5 segundoz behin).
  - top -o %CPU: Prozesuak automatikoki ordenatzen ditu CPU gehien kontsumitzen ari direnaren arabera.
  - top -u [erabiltzailea]: Erabiltzaile jakin baten prozesuak bakarrik erakusten ditu.

3. htop

- **Zer egiten du:** top komandoaren bertsio aurreratu eta bisualagoa da. Interfaze grafiko txiki bat ematen du terminalean, sagua erabili ahal izateko eta koloreekin laguntzeko.
- **Nola erabili:**
  - htop: Programa abiarazten du. Beheko aldean funtzio-teken zerrenda agertzen da (F1-etik F10-ra) laguntza, bilaketak, iragazkiak edo prozesuak hil ahal izateko.

4. man (Manual)

- **Zer egiten du:** Linuxeko komando bakoitzaren eskuliburu edo dokumentazio ofiziala erakusten du.
- **Nola erabili:**
  - man ps: ps komandoaren manuala irekitzen du.
  - man top / man htop: Komando horien argibide eta aukera guztiak erakusten ditu. Bertatik ateratzeko q sakatu behar da.

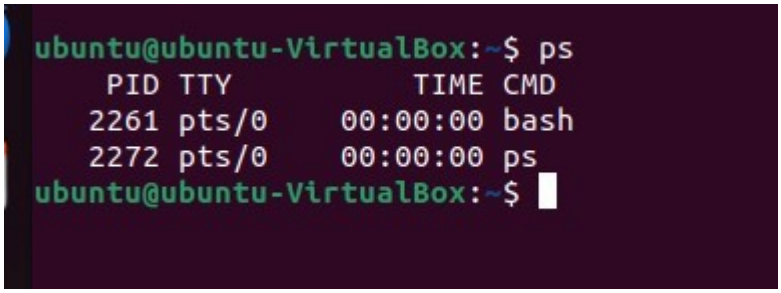
5. sudo su

- **Zer egiten du:** Erabiltzaile arrunt batetik (ubuntu) administratzaile edo root erabiltzaile izatera pasatzeko balio du.
- **Zergatik erabiltzen da:** Sistemako prozesu guztiak kudeatu, ikusi edo baimen bereziak eskatzen dituzten ekintzak egiteko (adibidez, beste erabiltzaile batzuen prozesuak ixteko).

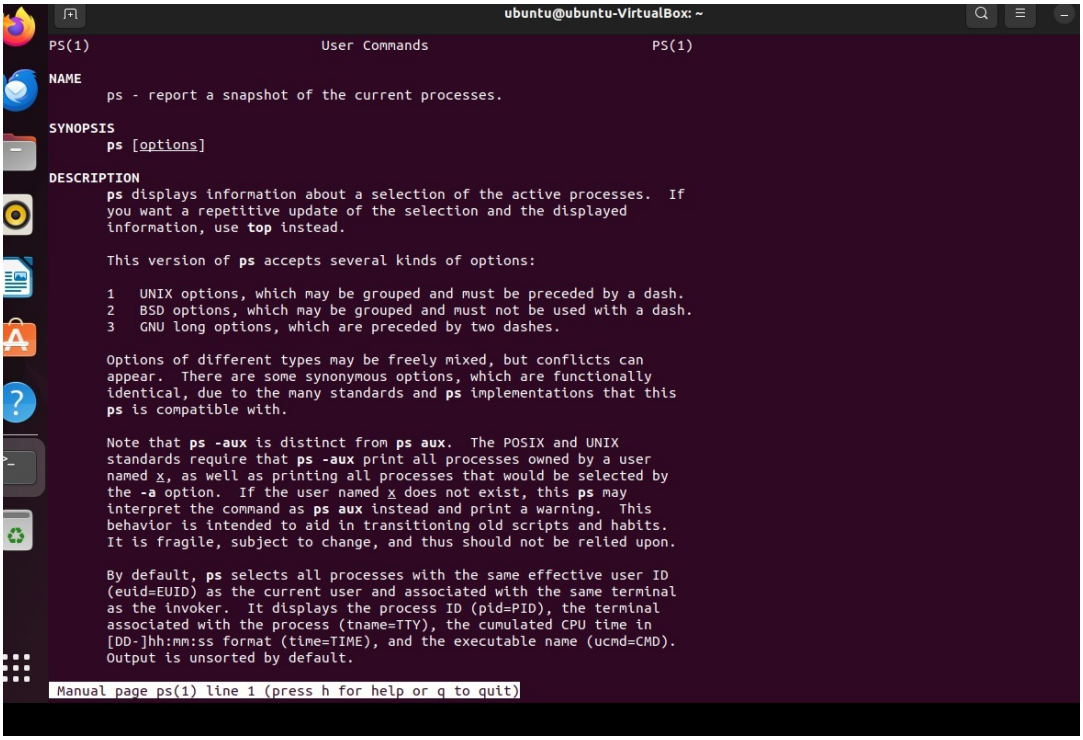
6. kill, killall eta Seinaleak

- **Zer egiten du (chatgpt egina):** Martxan dagoen prozesu bat gelditzeko edo ixteko erabiltzen dira, PID zenbakiaren edo izenaren bidez.
- **Nola erabili eta aldaerak:**
  - kill [PID]: Prozesuari amaitzeko eskatzen dio modu leunean (SIGTERM seinalea).
  - kill -9 [PID]: Prozesu bat **derrigorrez eta berehala** ixten du, sistema mailan (*SIGKILL* seinalea). Blokeatuta dauden aplikazioekin erabiltzen da.
  - kill -HUP [PID] edo kill -1: Prozesuari bere konfigurazioa berriro kargatzeko agintzen dio.
  - killall [izena]: Izen bera duten prozesu guztiak aldi berean ixten ditu (adibidez, killall firefox), PID zenbakia banaka bilatu behar izan gabe.

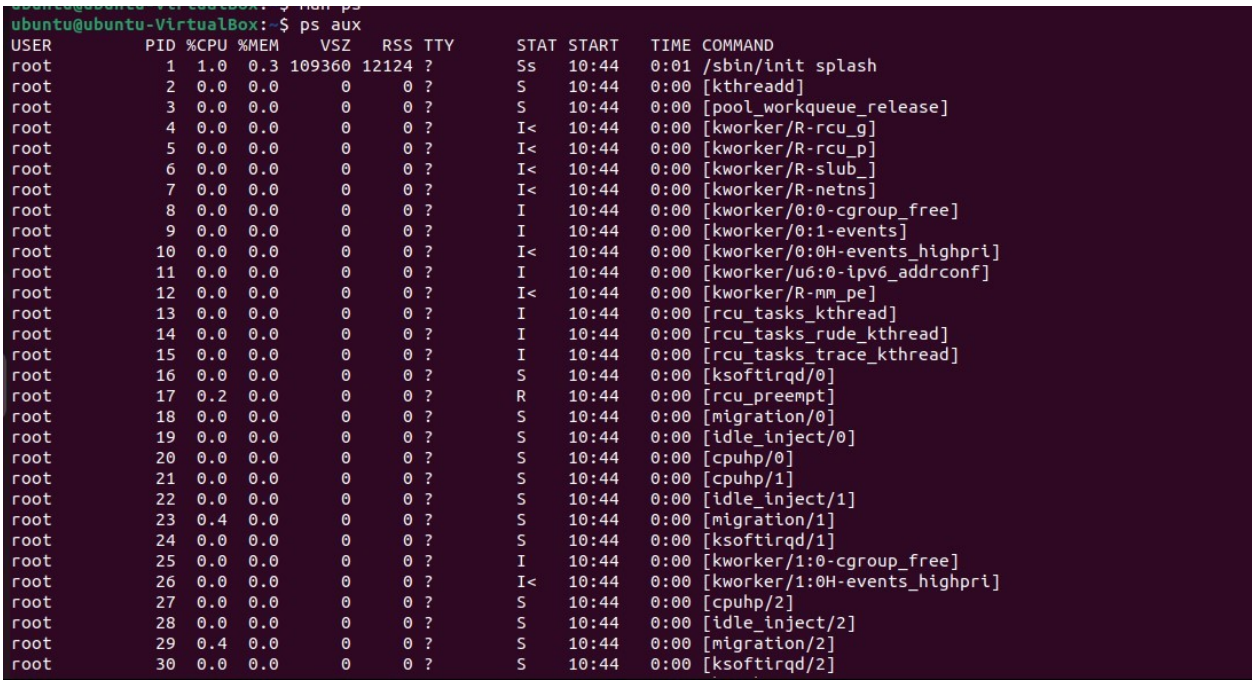
PS:



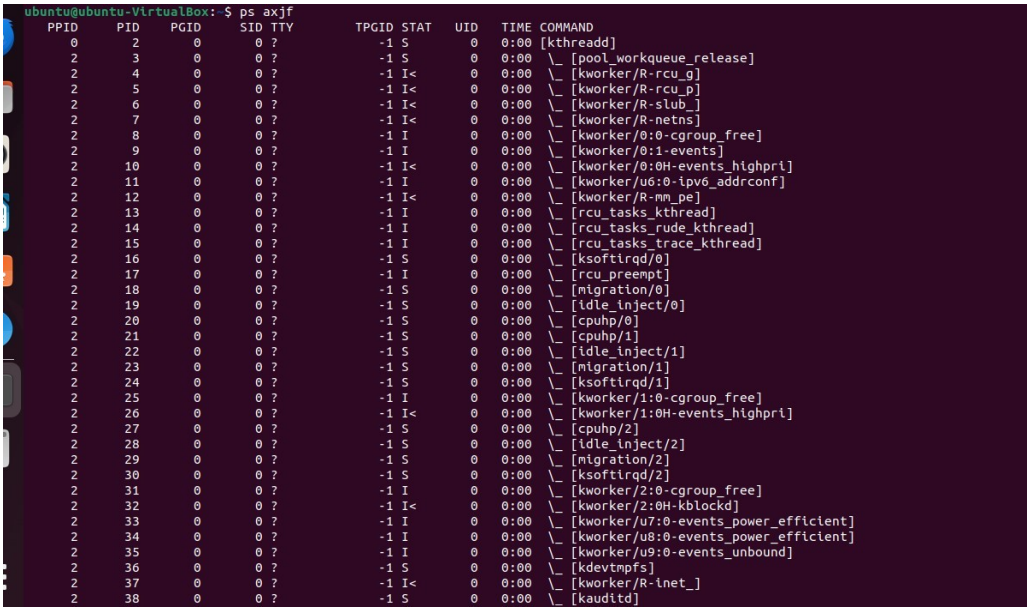
MAN PS:


ubuntu@ubuntu-VirtualBox: ~
PS(1) User Commands PS(1)
NAME
ps - report a snapshot of the current processes.
SYNOPSIS
ps [options]
DESCRIPTION
ps displays information about a selection of the active processes. If you want a repetitive update of the selection and the displayed information, use top instead.
This version of ps accepts several kinds of options:
1 UNIX options, which may be grouped and must be preceded by a dash.
2 BSD options, which may be grouped and must not be used with a dash.
3 GNU long options, which are preceded by two dashes.
Options of different types may be freely mixed, but conflicts can appear. There are some synonymous options, which are functionally identical, due to the many standards and ps implementations that this ps is compatible with.
Note that ps -aux is distinct from ps aux. The POSIX and UNIX standards require that ps -aux print all processes owned by a user named x, as well as printing all processes that would be selected by the -a option. If the user named x does not exist, this ps may interpret the command as ps aux instead and print a warning. This behavior is intended to aid in transitioning old scripts and habits. It is fragile, subject to change, and thus should not be relied upon.
By default, ps selects all processes with the same effective user ID (euid=EUID) as the current user and associated with the same terminal as the invoker. It displays the process ID (pid=PID), the terminal associated with the process (tname=TTY), the cumulated CPU time in [DD-]hh:mm:ss format (time=TIME), and the executable name (ucmd=CMD). Output is unsorted by default.
Manual page ps(1) line 1 (press h for help or q to quit)

PS AUX:

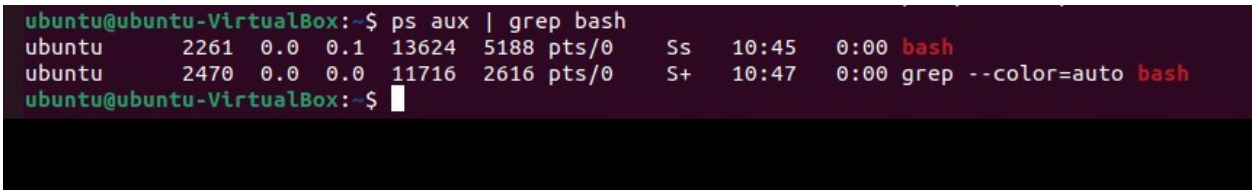

ubuntu@ubuntu-VirtualBox:~$ ps aux
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 1.0 0.3 109360 12124 ? Ss 10:44 0:01 /sbin/init splash
root 2 0.0 0.0 0 0 ? S 10:44 0:00 [kthreadd]
root 3 0.0 0.0 0 0 ? S 10:44 0:00 [pool_workqueue_release]
root 4 0.0 0.0 0 0 ? I< 10:44 0:00 [kworker/R-rcu_g]
root 5 0.0 0.0 0 0 ? I< 10:44 0:00 [kworker/R-rcu_p]
root 6 0.0 0.0 0 0 ? I< 10:44 0:00 [kworker/R-slub_]
root 7 0.0 0.0 0 0 ? I< 10:44 0:00 [kworker/R-netns]
root 8 0.0 0.0 0 0 ? I 10:44 0:00 [kworker/0:0-cgroup_free]
root 9 0.0 0.0 0 0 ? I 10:44 0:00 [kworker/0:1-events]
root 10 0.0 0.0 0 0 ? I< 10:44 0:00 [kworker/0:0H-events_highpri]
root 11 0.0 0.0 0 0 ? I 10:44 0:00 [kworker/u6:0-ipv6_addrconf]
root 12 0.0 0.0 0 0 ? I< 10:44 0:00 [kworker/R-mm_pe]
root 13 0.0 0.0 0 0 ? I 10:44 0:00 [rcu_tasks_kthread]
root 14 0.0 0.0 0 0 ? I 10:44 0:00 [rcu_tasks_rude_kthread]
root 15 0.0 0.0 0 0 ? I 10:44 0:00 [rcu_tasks_trace_kthread]
root 16 0.0 0.0 0 0 ? S 10:44 0:00 [ksoftirqd/0]
root 17 0.2 0.0 0 0 ? R 10:44 0:00 [rcu_preempt]
root 18 0.0 0.0 0 0 ? S 10:44 0:00 [migration/0]
root 19 0.0 0.0 0 0 ? S 10:44 0:00 [idle_inject/0]
root 20 0.0 0.0 0 0 ? S 10:44 0:00 [cpuhp/0]
root 21 0.0 0.0 0 0 ? S 10:44 0:00 [cpuhp/1]
root 22 0.0 0.0 0 0 ? S 10:44 0:00 [idle_inject/1]
root 23 0.4 0.0 0 0 ? S 10:44 0:00 [migration/1]
root 24 0.0 0.0 0 0 ? S 10:44 0:00 [ksoftirqd/1]
root 25 0.0 0.0 0 0 ? I 10:44 0:00 [kworker/1:0-cgroup_free]
root 26 0.0 0.0 0 0 ? I< 10:44 0:00 [kworker/1:0H-events_highpri]
root 27 0.0 0.0 0 0 ? S 10:44 0:00 [cpuhp/2]
root 28 0.0 0.0 0 0 ? S 10:44 0:00 [idle_inject/2]
root 29 0.4 0.0 0 0 ? S 10:44 0:00 [migration/2]
root 30 0.0 0.0 0 0 ? S 10:44 0:00 [ksoftirqd/2]

PS AXJF



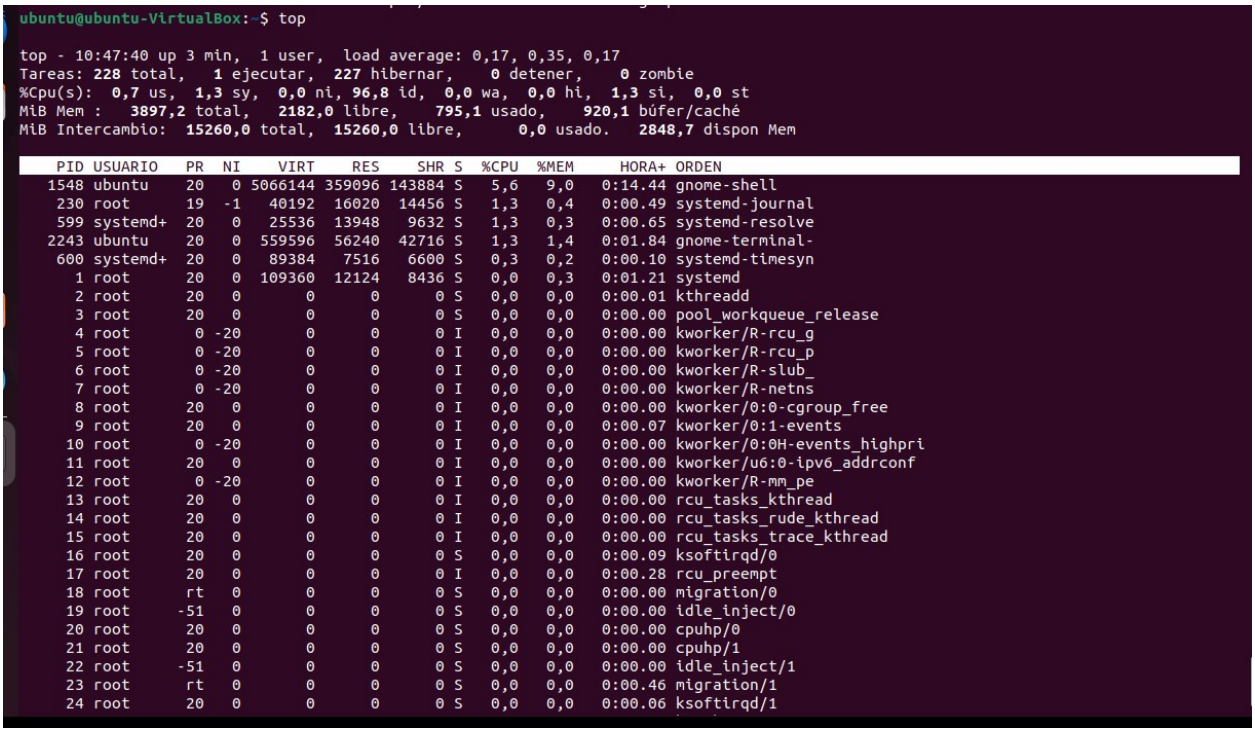
```
ubuntu@ubuntu-VirtualBox: $ ps axjf
   PPID     PID    PGID     SID TTY        TPGID STAT   UID   TIME COMMAND
      0       2       0       0 ?           -1 S        0   0:00 [kthreadd]
      2       3       0       0 ?           -1 S        0   0:00  \_ [pool_workqueue_release]
      2       4       0       0 ?           -1 I<       0   0:00  \_ [kworker/R-rcu_g]
      2       5       0       0 ?           -1 I<       0   0:00  \_ [kworker/R-rcu_p]
      2       6       0       0 ?           -1 I<       0   0:00  \_ [kworker/R-slub_]
      2       7       0       0 ?           -1 I<       0   0:00  \_ [kworker/R-netns]
      2       8       0       0 ?           -1 I        0   0:00  \_ [kworker/0:0-cgroup_free]
      2       9       0       0 ?           -1 I        0   0:00  \_ [kworker/0:1-events]
      2      10       0       0 ?           -1 I<       0   0:00  \_ [kworker/0:0H-events_highpri]
      2      11       0       0 ?           -1 I        0   0:00  \_ [kworker/u6:0-ipv6_addrconf]
      2      12       0       0 ?           -1 I<       0   0:00  \_ [kworker/R-mm_pe]
      2      13       0       0 ?           -1 I        0   0:00  \_ [rcu_tasks_kthread]
      2      14       0       0 ?           -1 I        0   0:00  \_ [rcu_tasks_rude_kthread]
      2      15       0       0 ?           -1 I        0   0:00  \_ [rcu_tasks_trace_kthread]
      2      16       0       0 ?           -1 S        0   0:00  \_ [ksoftirqd/0]
      2      17       0       0 ?           -1 I        0   0:00  \_ [rcu_preempt]
      2      18       0       0 ?           -1 S        0   0:00  \_ [migration/0]
      2      19       0       0 ?           -1 S        0   0:00  \_ [idle_inject/0]
      2      20       0       0 ?           -1 S        0   0:00  \_ [cpuhp/0]
      2      21       0       0 ?           -1 S        0   0:00  \_ [cpuhp/1]
      2      22       0       0 ?           -1 S        0   0:00  \_ [idle_inject/1]
      2      23       0       0 ?           -1 S        0   0:00  \_ [migration/1]
      2      24       0       0 ?           -1 S        0   0:00  \_ [ksoftirqd/1]
      2      25       0       0 ?           -1 I        0   0:00  \_ [kworker/1:0-cgroup_free]
      2      26       0       0 ?           -1 I<       0   0:00  \_ [kworker/1:0H-events_highpri]
      2      27       0       0 ?           -1 S        0   0:00  \_ [cpuhp/2]
      2      28       0       0 ?           -1 S        0   0:00  \_ [idle_inject/2]
      2      29       0       0 ?           -1 S        0   0:00  \_ [migration/2]
      2      30       0       0 ?           -1 S        0   0:00  \_ [ksoftirqd/2]
      2      31       0       0 ?           -1 I        0   0:00  \_ [kworker/2:0-cgroup_free]
      2      32       0       0 ?           -1 I<       0   0:00  \_ [kworker/2:0H-kblockd]
      2      33       0       0 ?           -1 I        0   0:00  \_ [kworker/u7:0-events_power_efficient]
      2      34       0       0 ?           -1 I        0   0:00  \_ [kworker/u8:0-events_power_efficient]
      2      35       0       0 ?           -1 I        0   0:00  \_ [kworker/u9:0-events_unbound]
      2      36       0       0 ?           -1 S        0   0:00  \_ [kdevtmpfs]
      2      37       0       0 ?           -1 I<       0   0:00  \_ [kworker/R-inet_]
      2      38       0       0 ?           -1 S        0   0:00  \_ [kauditd]
```

PS AUX | GREP DOCKER:



```
ubuntu@ubuntu-VirtualBox:~$ ps aux | grep bash
ubuntu      2261  0.0  0.1  13624  5188 pts/0    Ss   10:45   0:00 bash
ubuntu      2470  0.0  0.0  11716  2616 pts/0    S+   10:47   0:00 grep --color=auto bash
ubuntu@ubuntu-VirtualBox:~$
```

TOP:


ubuntu@ubuntu-VirtualBox:~$ top
top - 10:47:40 up 3 min, 1 user, load average: 0,17, 0,35, 0,17
Tareas: 228 total, 1 ejecutar, 227 hibernar, 0 detener, 0 zombie
%Cpu(s): 0,7 us, 1,3 sy, 0,0 ni, 96,8 id, 0,0 wa, 0,0 hi, 1,3 si, 0,0 st
MiB Mem : 3897,2 total, 2182,0 libre, 795,1 usado, 920,1 búfer/caché
MiB Intercambio: 15260,0 total, 15260,0 libre, 0,0 usado. 2848,7 dispon Mem
PID USUARIO PR NI VIRT RES SHR S %CPU %MEM HORA+ ORDEN
1548 ubuntu 20 0 5066144 359096 143884 S 5,6 9,0 0:14.44 gnome-shell
230 root 19 -1 40192 16020 14456 S 1,3 0,4 0:00.49 systemd-journal
599 systemd+ 20 0 25536 13948 9632 S 1,3 0,3 0:00.65 systemd-resolve
2243 ubuntu 20 0 559596 56240 42716 S 1,3 1,4 0:01.84 gnome-terminal-
600 systemd+ 20 0 89384 7516 6600 S 0,3 0,2 0:00.10 systemd-timesyn
1 root 20 0 109360 12124 8436 S 0,0 0,3 0:01.21 systemd
2 root 20 0 0 0 0 S 0,0 0,0 0:00.01 kthreadd
3 root 20 0 0 0 0 S 0,0 0,0 0:00.00 pool_workqueue_release
4 root 0 -20 0 0 0 I 0,0 0,0 0:00.00 kworker/R-rcu_g
5 root 0 -20 0 0 0 I 0,0 0,0 0:00.00 kworker/R-rcu_p
6 root 0 -20 0 0 0 I 0,0 0,0 0:00.00 kworker/R-slub_
7 root 0 -20 0 0 0 I 0,0 0,0 0:00.00 kworker/R-netns
8 root 20 0 0 0 0 I 0,0 0,0 0:00.00 kworker/0:0-cgroup_free
9 root 20 0 0 0 0 I 0,0 0,0 0:00.07 kworker/0:1-events
10 root 0 -20 0 0 0 I 0,0 0,0 0:00.00 kworker/0:0H-events_highpri
11 root 20 0 0 0 0 I 0,0 0,0 0:00.00 kworker/u6:0-ipv6_addrconf
12 root 0 -20 0 0 0 I 0,0 0,0 0:00.00 kworker/R-mm_pe
13 root 20 0 0 0 0 I 0,0 0,0 0:00.00 rcu_tasks_kthread
14 root 20 0 0 0 0 I 0,0 0,0 0:00.00 rcu_tasks_rude_kthread
15 root 20 0 0 0 0 I 0,0 0,0 0:00.00 rcu_tasks_trace_kthread
16 root 20 0 0 0 0 S 0,0 0,0 0:00.09 ksoftirqd/0
17 root 20 0 0 0 0 I 0,0 0,0 0:00.28 rcu_preempt
18 root rt 0 0 0 0 S 0,0 0,0 0:00.00 migration/0
19 root -51 0 0 0 0 S 0,0 0,0 0:00.00 idle_inject/0
20 root 20 0 0 0 0 S 0,0 0,0 0:00.00 cpuhp/0
21 root 20 0 0 0 0 S 0,0 0,0 0:00.00 cpuhp/1
22 root -51 0 0 0 0 S 0,0 0,0 0:00.00 idle_inject/1
23 root rt 0 0 0 0 S 0,0 0,0 0:00.46 migration/1
24 root 20 0 0 0 0 S 0,0 0,0 0:00.06 ksoftirqd/1

MAN TOP:

```
TOP(1)                                  User Commands                                  TOP(1)

NAME
       top - display Linux processes

SYNOPSIS
       top -hv|-bcEeHiOSs1 -d secs -n max -u|U user -p pids -o field -w [cols]

       The traditional switches `-' and whitespace are optional.

DESCRIPTION
       The top program provides a dynamic real-time view of a running system.  It can display system summary information as well
       as a list of processes or threads currently being managed by the Linux kernel.  The types of system  summary  information
       shown  and  the  types,  order  and  size  of  information  displayed  for  processes  are all user configurable and that
       configuration can be made persistent across restarts.

       The program provides a limited interactive interface for process manipulation as well as a much more extensive  interface
       for personal configuration  --  encompassing every aspect of its operation.  And while top is referred to throughout this
       document, you are free to name the program anything you wish.  That new name, possibly an alias, will then  be  reflected
       on top's display and used when reading and writing a configuration file.

OVERVIEW
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           OVERVIEW
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              c. MEMORY Usage
           3. FIELDS / Columns Display
              a. DESCRIPTIONS of Fields
              b. MANAGING Fields
           4. INTERACTIVE Commands
              a. GLOBAL Commands
              b. SUMMARY AREA Commands
 Manual page top(1) line 1 (press h for help or q to quit)
```

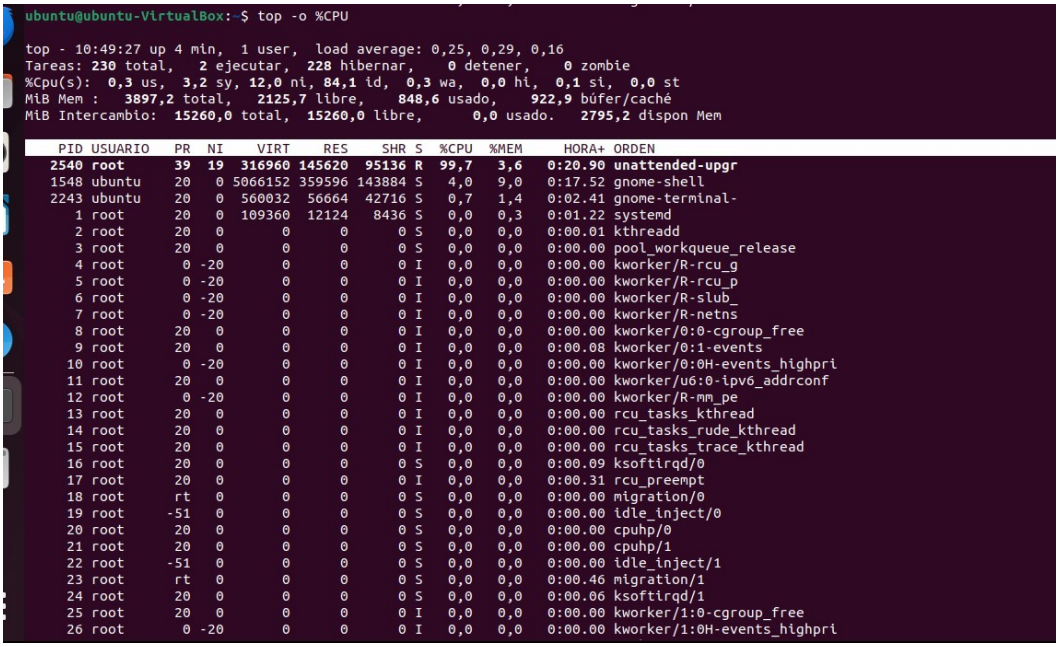
TOP D -5:

```
ubuntu@ubuntu-VirtualBox:~$ top -d 5

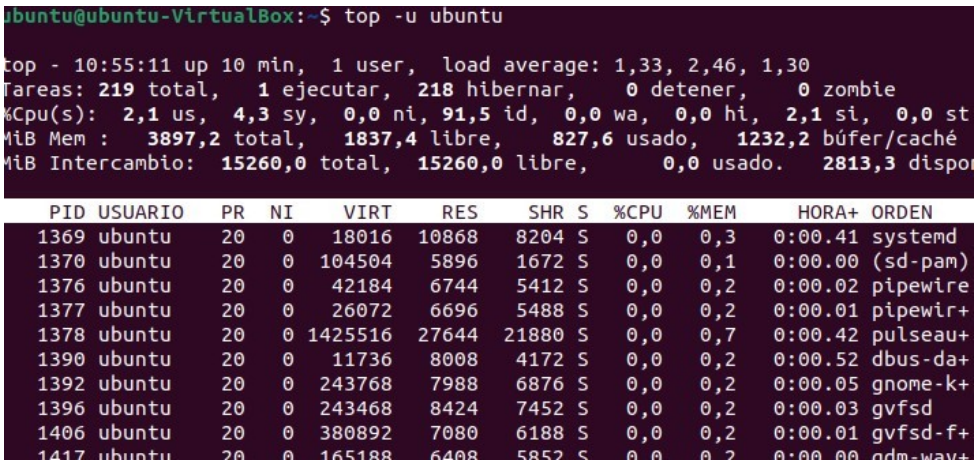
top - 10:48:57 up 4 min,  1 user,  load average: 0,05, 0,27, 0,15
Tareas: 227 total,   2 ejecutar,  225 hibernar,    0 detener,    0 zombie
%Cpu(s):  0,0 us,  1,5 sy,  0,0 ni, 98,5 id,  0,0 wa,  0,0 hi,  0,0 si,  0,0 st
MiB Mem :   3897,2 total,   2181,0 libre,    795,5 usado,    920,6 búfer/caché
MiB Intercambio:  15260,0 total,  15260,0 libre,      0,0 usado.   2848,3 dispon Mem

    PID USUARIO   PR  NI    VIRT    RES    SHR S  %CPU  %MEM     HORA+ ORDEN
   1548 ubuntu    20   0 5066136 359636 143884 S   8,3   9,0   0:16.81 gnome-shell
      1 root      20   0  109360  12124   8436 S   0,0   0,3   0:01.21 systemd
      2 root      20   0       0      0      0 S   0,0   0,0   0:00.01 kthreadd
      3 root      20   0       0      0      0 S   0,0   0,0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0,0   0,0   0:00.00 kworker/R-rcu_g
      5 root       0 -20       0      0      0 I   0,0   0,0   0:00.00 kworker/R-rcu_p
      6 root       0 -20       0      0      0 I   0,0   0,0   0:00.00 kworker/R-slub_
      7 root       0 -20       0      0      0 I   0,0   0,0   0:00.00 kworker/R-netns
      8 root      20   0       0      0      0 I   0,0   0,0   0:00.00 kworker/0:0-cgroup_free
      9 root      20   0       0      0      0 I   0,0   0,0   0:00.08 kworker/0:1-events
     10 root       0 -20       0      0      0 I   0,0   0,0   0:00.00 kworker/0:0H-events_highpri
     11 root      20   0       0      0      0 I   0,0   0,0   0:00.00 kworker/u6:0-ipv6_addrconf
     12 root       0 -20       0      0      0 I   0,0   0,0   0:00.00 kworker/R-mm_pe
     13 root      20   0       0      0      0 I   0,0   0,0   0:00.00 rcu_tasks_kthread
     14 root      20   0       0      0      0 I   0,0   0,0   0:00.00 rcu_tasks_rude_kthread
     15 root      20   0       0      0      0 I   0,0   0,0   0:00.00 rcu_tasks_trace_kthread
     16 root      20   0       0      0      0 S   0,0   0,0   0:00.09 ksoftirqd/0
     17 root      20   0       0      0      0 I   0,0   0,0   0:00.30 rcu_preempt
     18 root      rt   0       0      0      0 S   0,0   0,0   0:00.00 migration/0
     19 root      51   0       0      0      0 S   0,0   0,0   0:00.00 idle_inject/0
```

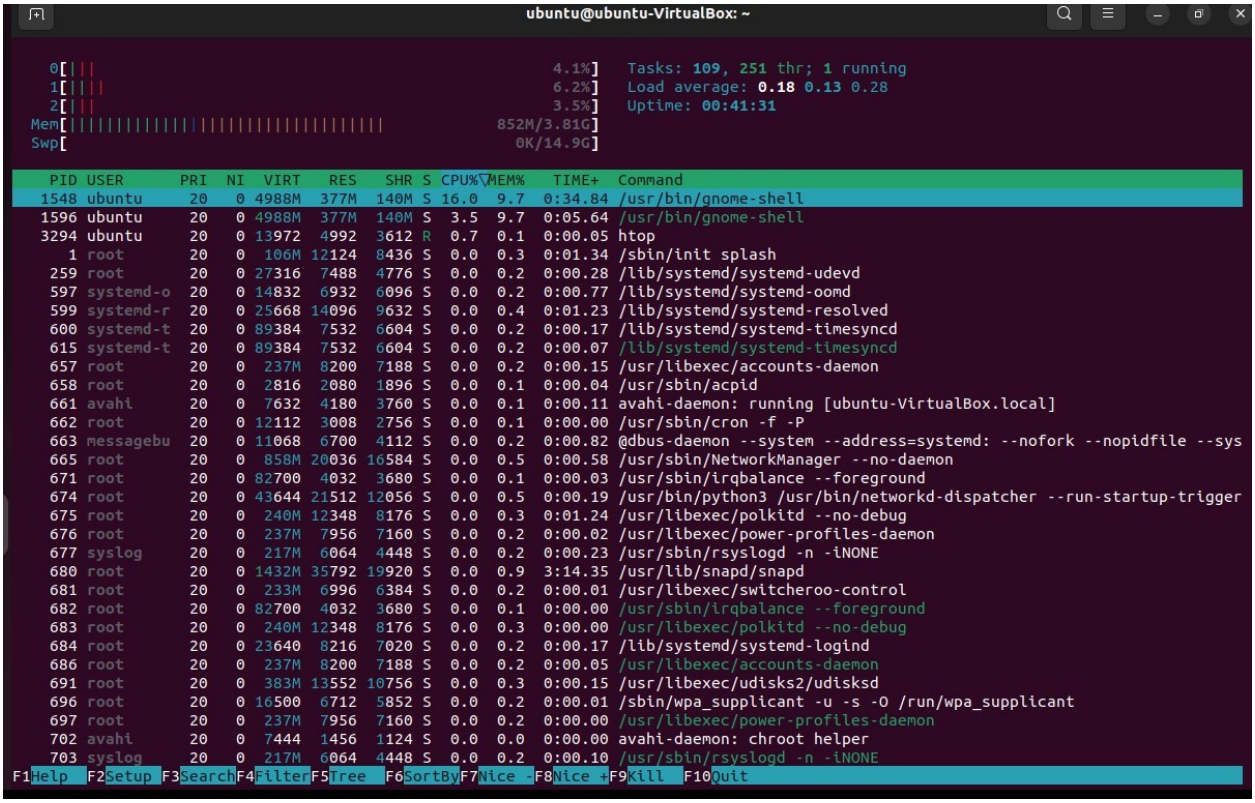
TOP -O %CPU:



TOP -U UBUNTU (erabiltzailearen izena):



HTOP:


ubuntu@ubuntu-VirtualBox: ~
0[4.1%] Tasks: 109, 251 thr; 1 running
1[6.2%] Load average: 0.18 0.13 0.28
2[3.5%] Uptime: 00:41:31
Mem[852M/3.81G]
Swp[0K/14.9G]
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1548 ubuntu 20 0 4988M 377M 140M S 16.0 9.7 0:34.84 /usr/bin/gnome-shell
1596 ubuntu 20 0 4988M 377M 140M S 3.5 9.7 0:05.64 /usr/bin/gnome-shell
3294 ubuntu 20 0 13972 4992 3612 R 0.7 0.1 0:00.05 htop
1 root 20 0 106M 12124 8436 S 0.0 0.3 0:01.34 /sbin/init splash
259 root 20 0 27316 7488 4776 S 0.0 0.2 0:00.28 /lib/systemd/systemd-udevd
597 systemd-o 20 0 14832 6932 6096 S 0.0 0.2 0:00.77 /lib/systemd/systemd-oomd
599 systemd-r 20 0 25668 14096 9632 S 0.0 0.4 0:01.23 /lib/systemd/systemd-resolved
600 systemd-t 20 0 89384 7532 6604 S 0.0 0.2 0:00.17 /lib/systemd/systemd-timesyncd
615 systemd-t 20 0 89384 7532 6604 S 0.0 0.2 0:00.07 /lib/systemd/systemd-timesyncd
657 root 20 0 237M 8200 7188 S 0.0 0.2 0:00.15 /usr/libexec/accounts-daemon
658 root 20 0 2816 2080 1896 S 0.0 0.1 0:00.04 /usr/sbin/acpid
661 avahi 20 0 7632 4180 3760 S 0.0 0.1 0:00.11 avahi-daemon: running [ubuntu-VirtualBox.local]
662 root 20 0 12112 3008 2756 S 0.0 0.1 0:00.00 /usr/sbin/cron -f -P
663 messagebu 20 0 11068 6700 4112 S 0.0 0.2 0:00.82 @dbus-daemon --system --address=systemd: --nofork --nopidfile --sys
665 root 20 0 858M 20036 16584 S 0.0 0.5 0:00.58 /usr/sbin/NetworkManager --no-daemon
671 root 20 0 82700 4032 3680 S 0.0 0.1 0:00.03 /usr/sbin/irqbalance --foreground
674 root 20 0 43644 21512 12056 S 0.0 0.5 0:00.19 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-trigger
675 root 20 0 240M 12348 8176 S 0.0 0.3 0:01.24 /usr/libexec/polkitd --no-debug
676 root 20 0 237M 7956 7160 S 0.0 0.2 0:00.02 /usr/libexec/power-profiles-daemon
677 syslog 20 0 217M 6064 4448 S 0.0 0.2 0:00.23 /usr/sbin/rsyslogd -n -iNONE
680 root 20 0 1432M 35792 19920 S 0.0 0.9 3:14.35 /usr/lib/snapd/snapd
681 root 20 0 233M 6996 6384 S 0.0 0.2 0:00.01 /usr/libexec/switcheroo-control
682 root 20 0 82700 4032 3680 S 0.0 0.1 0:00.00 /usr/sbin/irqbalance --foreground
683 root 20 0 240M 12348 8176 S 0.0 0.3 0:00.00 /usr/libexec/polkitd --no-debug
684 root 20 0 23640 8216 7020 S 0.0 0.2 0:00.17 /lib/systemd/systemd-logind
686 root 20 0 237M 8200 7188 S 0.0 0.2 0:00.05 /usr/libexec/accounts-daemon
691 root 20 0 383M 13552 10756 S 0.0 0.3 0:00.15 /usr/libexec/udisks2/udisksd
696 root 20 0 16500 6712 5852 S 0.0 0.2 0:00.01 /sbin/wpa_supplicant -u -s -O /run/wpa_supplicant
697 root 20 0 237M 7956 7160 S 0.0 0.2 0:00.00 /usr/libexec/power-profiles-daemon
702 avahi 20 0 7444 1456 1124 S 0.0 0.0 0:00.00 avahi-daemon: chroot helper
703 syslog 20 0 217M 6064 4448 S 0.0 0.2 0:00.10 /usr/sbin/rsyslogd -n -iNONE
F1Help F2Setup F3SearchF4FilterF5Tree F6SortByF7Nice -F8Nice +F9Kill F10Quit

MAN TOP:


ubuntu@ubuntu-VirtualBox: ~
HTOP(1) User Commands HTOP(1)
NAME
htop - interactive process viewer
SYNOPSIS
htop [-dCFhpustvH]
DESCRIPTION
htop is a cross-platform ncurses-based process viewer.
It is similar to top, but allows you to scroll vertically and horizontally, and interact using a pointing device (mouse).
You can observe all processes running on the system, along with their command line arguments, as well as view them in a
tree format, select multiple processes and acting on them all at once.
Tasks related to processes (killing, renicing) can be done without entering their PIDs.
COMMAND-LINE OPTIONS
Mandatory arguments to long options are mandatory for short options too.
-d --delay=DELAY
Delay between updates, in tenths of seconds. If the delay value is less than 1 it is increased to 1, i.e. 1/10
second. If the delay value is greater than 100, it is decreased to 100, i.e. 10 seconds.
-C --no-color --no-colour
Start htop in monochrome mode
-F --filter=FILTER
Filter processes by command
-h --help
Display a help message and exit
-p --pid=PID,PID...
Show only the given PIDs
-s --sort-key COLUMN
Sort by this column (use --sort-key help for a column list). This will force a list view unless you specify -t at
the same time.
Manual page htop(1) line 1 (press h for help or q to quit)

KILL (ID PROZESUA):

```
ubuntu@ubuntu-VirtualBox:~$ ps -aux | grep docker
ubuntu      3322  0.0  0.0  11716  2612 pts/0    S+   11:27   0:00 grep --color=auto docker
ubuntu@ubuntu-VirtualBox:~$ kill 3322
bash: kill: (3322) - No existe el proceso
ubuntu@ubuntu-VirtualBox:~$
```

kill -KILL (PID PROZESUA):

```
ubuntu@ubuntu-VirtualBox:~$ sudo su
root@ubuntu-VirtualBox:/home/ubuntu# cd
root@ubuntu-VirtualBox:~# ps aux |grep docker
root        3464  0.0  0.0  11716  2608 pts/1    R+   11:32   0:00 grep --color=auto docker
root@ubuntu-VirtualBox:~#
```

KILL -9 (ID PROZESUA):

```
root@ubuntu-VirtualBox:~# kill -9 3464
bash: kill: (3464) - No existe el proceso
```

kill -KILL (ID PROZESUA):

```
root@ubuntu-VirtualBox:~# kill -KILL 3364
bash: kill: (3364) - No existe el proceso
```

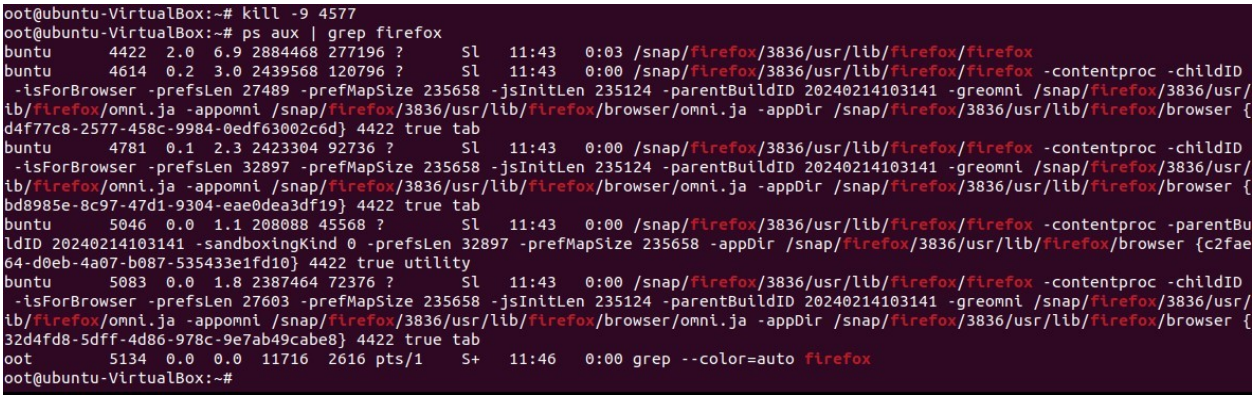
KILL -HUP (ID APACHE (firefox)):

```
root@ubuntu-VirtualBox:~# kill -HUP 5066
root@ubuntu-VirtualBox:~# kill -1 5048
root@ubuntu-VirtualBox:~# kill -9 3484
bash: kill: (3484) - No existe el proceso
root@ubuntu-VirtualBox:~# ps aux | grep firefox
ubuntu      4422  3.1  7.1 2892872 283572 ?      Sl   11:43   0:03 /snap/firefox/3836/usr/lib/firefox/firefox
ubuntu      4577  0.0  1.2 209312 48804 ?        Sl   11:43   0:00 /snap/firefox/3836/usr/lib/firefox/firefox -contentproc -parentB
ildID 20240214103141 -prefsLen 27349 -prefMapSize 235658 -appDir /snap/firefox/3836/usr/lib/firefox/browser {3b4090c7-c051-40ee-b51
-514ba9684ef7} 4422 true socket
ubuntu      4614  0.3  3.0 2440464 121692 ?      Sl   11:43   0:00 /snap/firefox/3836/usr/lib/firefox/firefox -contentproc -childID
1 -isForBrowser -prefsLen 27489 -prefMapSize 235658 -jsInitLen 235124 -parentBuildID 20240214103141 -greomni /snap/firefox/3836/usr
lib/firefox/omni.ja -appomni /snap/firefox/3836/usr/lib/firefox/browser/omni.ja -appDir /snap/firefox/3836/usr/lib/firefox/browser
4d4f77c8-2577-458c-9984-0edf63002c6d} 4422 true tab
ubuntu      4781  0.2  2.3 2423304 92736 ?       Sl   11:43   0:00 /snap/firefox/3836/usr/lib/firefox/firefox -contentproc -childID
2 -isForBrowser -prefsLen 32897 -prefMapSize 235658 -jsInitLen 235124 -parentBuildID 20240214103141 -greomni /snap/firefox/3836/usr
lib/firefox/omni.ja -appomni /snap/firefox/3836/usr/lib/firefox/browser/omni.ja -appDir /snap/firefox/3836/usr/lib/firefox/browser
abd8985e-8c97-47d1-9304-eae0dea3df19} 4422 true tab
ubuntu      5046  0.0  1.1 208088 45568 ?        Sl   11:43   0:00 /snap/firefox/3836/usr/lib/firefox/firefox -contentproc -parentB
ildID 20240214103141 -sandboxingKind 0 -prefsLen 32897 -prefMapSize 235658 -appDir /snap/firefox/3836/usr/lib/firefox/browser {c2fa
064-d0eb-4a07-b087-535433e1fd10} 4422 true utility
ubuntu      5083  0.0  1.8 2387464 72376 ?       Sl   11:43   0:00 /snap/firefox/3836/usr/lib/firefox/firefox -contentproc -childID
5 -isForBrowser -prefsLen 27603 -prefMapSize 235658 -jsInitLen 235124 -parentBuildID 20240214103141 -greomni /snap/firefox/3836/usr
lib/firefox/omni.ja -appomni /snap/firefox/3836/usr/lib/firefox/browser/omni.ja -appDir /snap/firefox/3836/usr/lib/firefox/browser
e32d4fd8-5dff-4d86-978c-9e7ab49cabe8} 4422 true tab
```

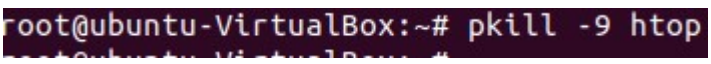
KILL -1 (ID APACHE (firefox)):



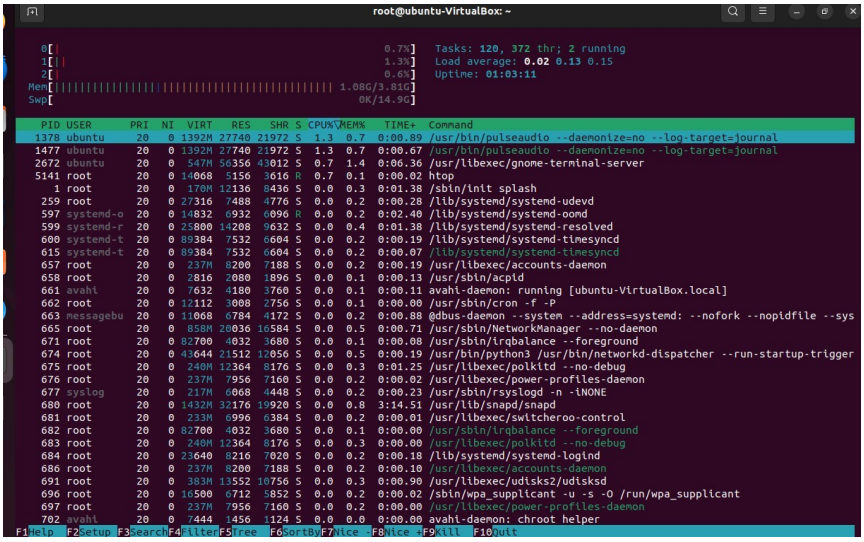
KILL -9 (ID APACHE (firefox)):



PKILL -9 HTOP:



HTOP:



KILLALL (FIREFOX):

